SIFT Feature Matching

Under Real-World Conditions

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Dataset & Research Question

Research Question

Which real-world condition most degrades SIFT feature matching, and why do some landmarks fail completely?

Metric: RANSAC Inlier Count

Green lines = inliers
(descriptor match + geometric check passed)

Red lines = outliers (rejected by RANSAC)

Dataset

Paris Buildings Dataset (Flickr tourist photos)
4 landmarks: Eiffel, Louvre, Notre Dame, Sacré-Cœur
36 image pairs, hand-labeled by visual inspection

Easy

Similar angle & lighting

Medium

Different angle or lighting

Hard

Night/day, crowds, extreme viewpoint

Stress

Maximally degraded pair

Pipeline

1	2	3	4	5
Load pair	Grayscale + Blur (3×3)	SIFT Keypoints	Lowe's Ratio Test (0.70)	RANSAC → Inliers

Grid search over 48 combinations
Selected: **blur=(3,3) · ratio=0.70 · reproj=8.0 px**

Ratio 0.70 (stricter than Lowe 2004 default 0.75) gave RANSAC cleaner input and unlocked the first successful Louvre match.

Green lines = inliers **Red lines** = outliers

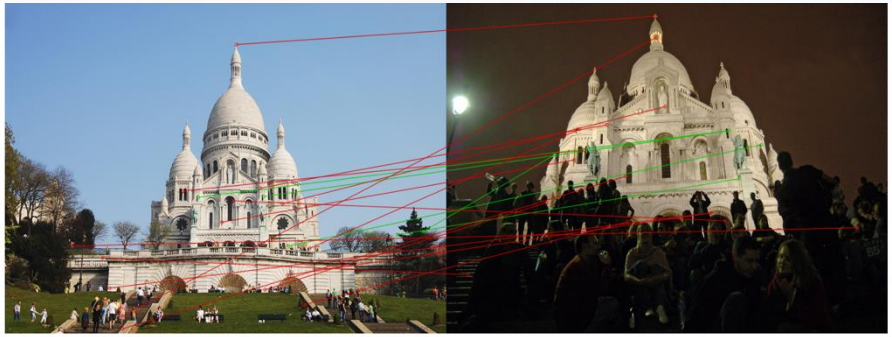
Sacré-Cœur easy pair: 256 inliers

Easy vs Hard: Sacré-Cœur

Easy: 256 inliers

Dense green lines connect matching keypoints that passed both Lowe's ratio test and RANSAC geometric verification.

Hard: 5 inliers



Almost all lines are red: 5 inliers from a night-time, crowd-heavy viewpoint.
Same landmark. 98% fewer verified matches.

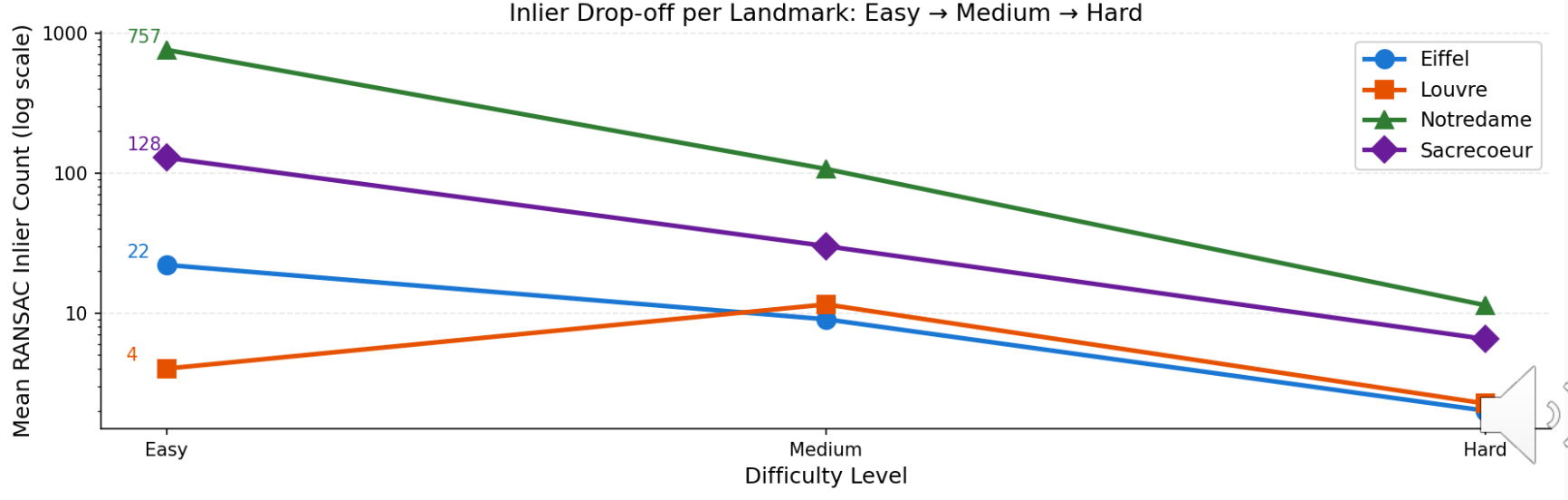
Why Do Some Landmarks Fail?

Notre Dame & Sacré-Cœur: succeed

Dense corners, edges & texture transitions → unique 128-D descriptors
Best match distance clearly beats second-best → Lowe's ratio test passes
Constrained shooting positions → stable surrounding context

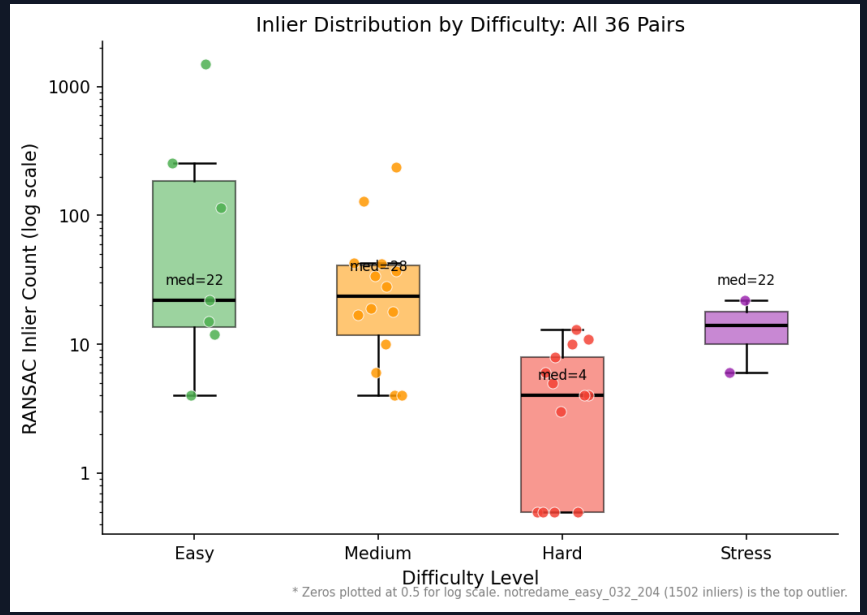
Louvre: nearly fails at every level

Identical windows & columns → same gradient histogram repeats
Best & second-best match distances nearly equal → ratio test rejects almost everything
Medium avg > Easy avg. Difficulty label has no predictive power.



Results

Notre Dame	Easy→Med→Hard: 757 → 107 → 11	**98% drop**
Sacré-Cœur	Easy→Med→Hard: 128 → 30 → 6.5	**95% drop**
Eiffel	Easy→Med→Hard: 22 → 9 → 2	**91% drop**
Louvre	Easy→Med→Hard: 4 → 11.5 → 2.3	**flat, label meaningless**



Key Finding

Building surface + shooting context determines SIFT performance more than the difficulty label.

Louvre nearly fails at every level. Notre Dame drops 98%. Same algorithm, same parameters.

Thank You

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